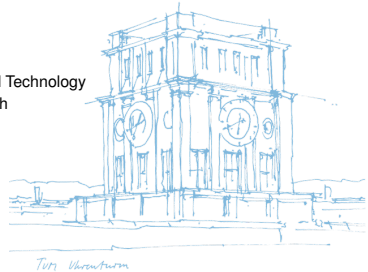


An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

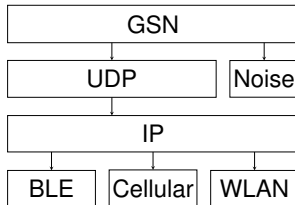
Dan Bachar, David Guzman

Monday 18th May, 2026

Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

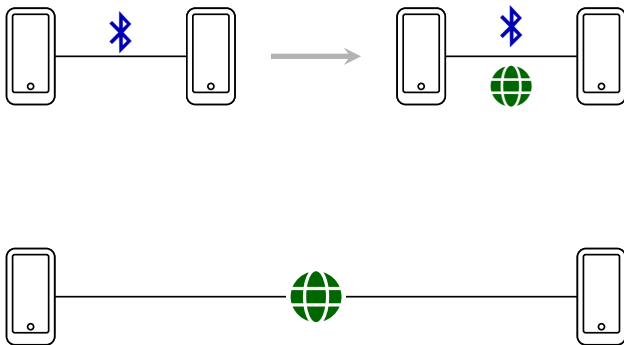


Architecture



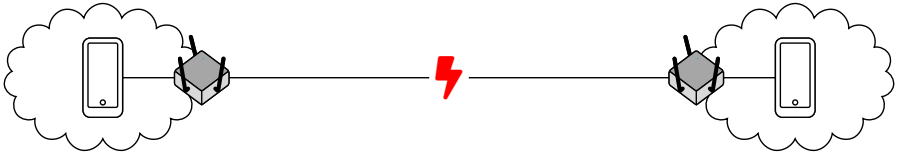
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

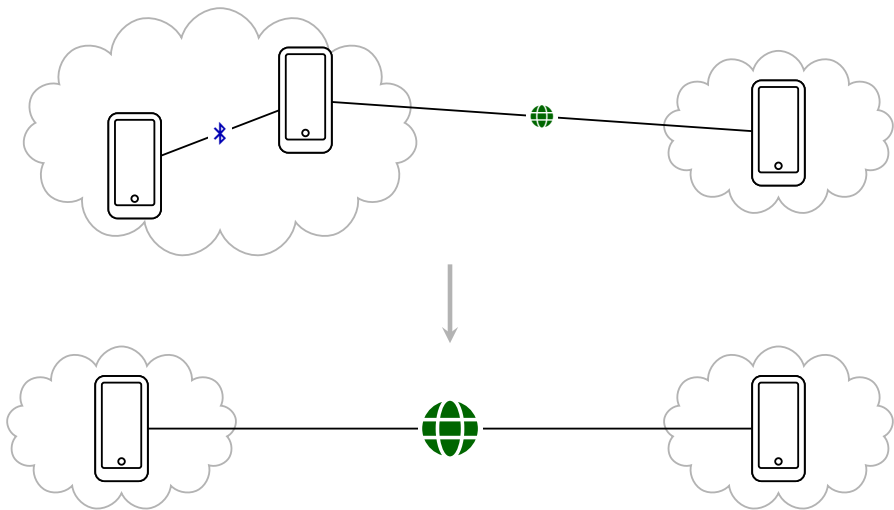


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

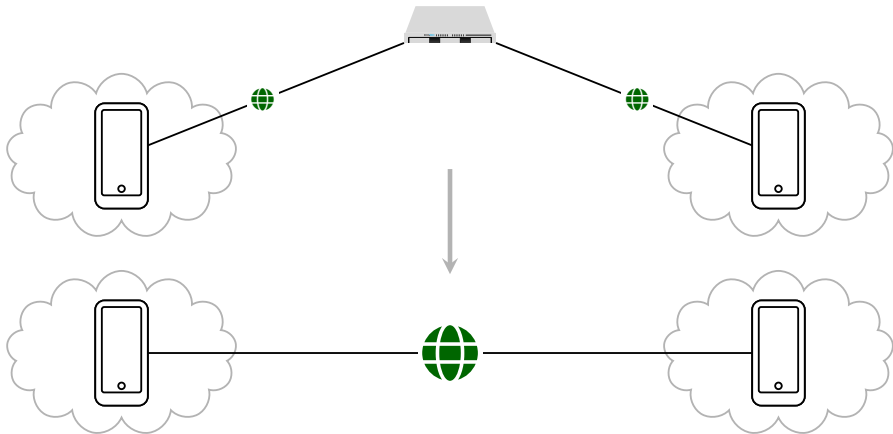
NAT



Well-connected friends



Rendezvous Server



Since mid-April: Flutter project work

- Built the transport around cryptographic identity: Ed25519 public keys as peer identity and signed packets as the basic trust object.
- Implemented BLE discovery and nearby messaging, including configurable BLE central/peripheral roles for device testing.
- Added UDP/UDX connectivity plumbing for longer-range friend communication and an Internet-scale baseline.
- Added NAT traversal support: public address discovery, well-connected friend facilitators, hole-punching state, reconnect signaling, and rendezvous coordination.
- Added friendship state in Redux: pending/accepted friendships, persistence, unfriend handling, friends-of-friends, and friend-shared rendezvous servers.
- Added Noise sessions for hop-level channel security between connected peers.
- Added delivery machinery: ACK/read receipts, fragmentation/reassembly for larger payloads, duplicate suppression, and routing tests.

Mobility model: city-scale BLE relay

Goal

- Test if BLE relay can replace UDP locally.
- Compare direct, friend, and open relay (simulated).
- Report delivery, latency, overhead, and abuse surface.

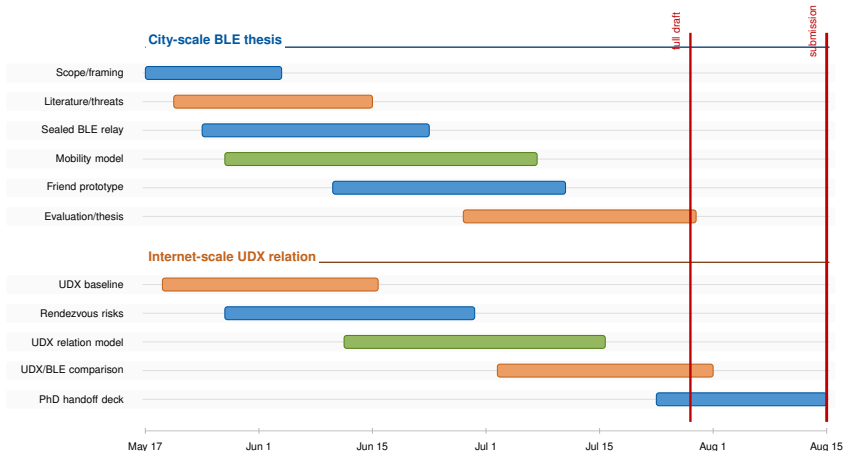
What we track

- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost: relay bytes, duplicates, queues, scan/advertise time, battery proxy.

How

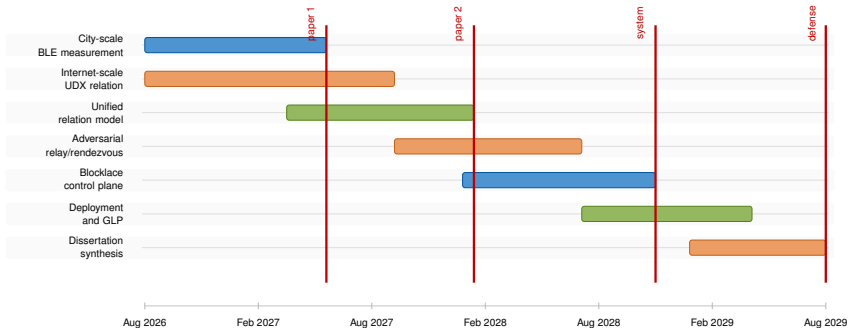
- Encrypted local trace log.
- Upload only by explicit opt-in.
- Replay traces in simulator calibrated by BLE contacts.
- No content; coarse location only by separate opt-in.

Master gantt: thesis plus UDX track



- Thesis core: BLE mobility and secure local relay; Keidar–Shapiro thread: keep UDX as an Internet-scale relation model and comparison.

PhD runway: two relation systems



- Thesis-to-PhD bridge: compare what grassroots relations mean under local physical mobility and under Internet-scale UDX connectivity.